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T^3 Talk2Text – A model for near real-time voice transcription in virtual group meetings

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Abstract
Group projects are an important part of many educational programs. In these projects, groupware tools like shared workspaces, shared editors, and synchronous or asynchronous communication tools (video conferencing, chat, email, forum) are often utilized. For synchronous collaboration, video conferencing systems are often used since they allow for direct and effortless informal communication via speech and video. If the group is to reflect about the content and way of communication and collaboration within and across meetings, the group needs access to the development of artifacts as well as the conversation of the meetings. While some video conferencing systems allow recording of a meeting, accessing content and dialogue structure requires real-time viewing and detailed note-taking. To reduce this effort, a video chat should include a transcription functionality that is able to support students in an online group work session by providing a video chat with automatic creation of a transcript. For this purpose, a video communication tool named T^3 was developed that enables communication between learners and generates a transcript through an integrated automatic speech recognition (ASR) system. The transcript can be used to reflect on the conversations in order to recall the topics discussed or to identify group work problems, such as insufficient participation, coordination and collaboration problems. The implementation of T^3 uses voice activation detection, WebRTC and ASR models to maintain a high level of quality in the transcription process. Initial functionality tests demonstrated the ability of T^3 to create accurate group discussion transcripts, making it attractive for students and teachers to assess and improve communication and collaboration, and for researchers studying group discussions.

Keywords Audio/video conferencing · Audio to text · Group transcription · ASR · Automatic speech recognition · e-learning

1 Introduction

In education, taking notes is a common method for improving the learning process. Many studies describe this fact and try to make the positive effects of note-taking more accessible through various methods and media [1–3]. These positive effects can be utilized not only in traditional learning environments but also in virtual environments [4]. Nowadays, virtual meetings and remote work are no longer an exception. Even after COVID-19, learning groups at universities often work remotely, for example, through shared workspaces, collaborative writing tools or virtual communication tools, to achieve their learning goals. While working together, problems can arise such as unequal participation, a lack of responsibility on the part of individual participants, a lack of orientation and a lack of or conflictual communication

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[5–10]. To minimize these problems and prevent them from occurring in the future, it is important to analyze the group work process and identify opportunities for improvement [11, 12]. However, this requires a documentation of the group work process including the development of artifacts, such as shared documents, and concurrent dialogues/discussion. This could be achieved, for example, through logs and note-taking. With the various information channels that can be used to communicate in a group, including verbal, non-verbal and written communication, the effort required to record and analyze the content can vary. In contrast to written communication such as emails or chats, it is often very difficult and time-consuming to collect, store and extract information from verbal communication [13], so that this process is rarely carried out. However, this is important to give group members the opportunity to reflect on conversations and remember issues that have been clarified [14]. Though it is possible to record the meeting and then analyze the video, it is very cumbersome and requires a lot of cognitive effort and time. A simplification of logging verbal communication can be achieved through the use of an automatic speech recognition (ASR) system. Such a system can be started by the individual members of the group to record what they themselves have said and generate an automatic transcription. The influence of transcribing from speech to text in research institutions has already been well studied. Hwang et al. [15], Kuo et al. [16] and Huang et al. [17] examined the effect of speech to text tools on synchronous online teaching and were able to find significant positive effects. A positive influence of transcription tools on teaching was found not only among online students, but also among students with cognitive or physical disabilities, non-native speakers, and students in traditional learning environments [18]. However, one disadvantage is that only what is said by one person is recorded and it is always necessary to exchange the respective transcripts among group members. This also makes it more difficult to construct the history of the conversation from several transcripts. Alternatively, the entire conversation could be recorded and transcribed, but then information about the respective speaker and the reference of the speaker to previous contributions would be lost.

To support reflection, all group members should have access to an up-to-date transcript of the group's conversation, including information on the respective speaker. This can be achieved by (1) using an ASR tool for capturing contributions of group members and (2) compiling the results into a transcript of the entire conversation.

Similar functionalities already exist as commercial solutions, such as Microsoft Teams live transcription [19]. However, they require cost-intensive licenses and imply restrictions on use. For example, Microsoft Teams live transcription can only be used in the desktop app by special users. Also, the data protection offered by Microsoft [20] is unsatisfactory for some users when it comes to sensitive data.

For effective student support and the broad use of such a tool in teaching, it is necessary to offer a data-secure open-source alternative avoiding high costs. To develop such a system that supports the needs of students in a collaborative learning environment, the following research questions arise:

- RQ 1: What are the requirements for a system that supports both, video communication between students working in groups and reflection based on the transcription of the entire conversation in a group?
- RQ 2: How can a video communication and transcription system be constructed that meets these requirements?

2 Audio/video conferences and group transcripts in learning groups

As discussed in the introduction, communication is an important part of the group work process in learning groups. If communication is missing or incorrect, it can lead to poor results in group work and learning [5]. In virtual or remote meetings students need to communicate via synchronous computer-mediated communication channels such as chat, comments in shared editors, or audio and video channels. Audio and video carry a lot of information content that is important for information exchange in a group [21], both in co-located and distributed meetings.

To support students in their group work and help identify potential problems early on, communication within the group may be analyzed by the students themselves, by teachers to assess quality and problems of communication within a group, or researchers to assess quality and problems of communication of many groups. As a prerequisite for any analysis, data about the group work process and communication, including those using audio and video channels, must be captured and made accessible. In case of a distributed group using an audio/video conference tool, the audio and video streams during a running conference must be accessible so that a transcript of the group communication may be produced. Data privacy regulations applicable to educational settings, quality demands of later analysis methods, and easy accessibility of the transcript suggest the development of a tool integrating video communication and transcription, which can be made available to students, teachers and researchers within an educational institution.

Students who may join a conference late (i.e. latecomers) may benefit from such a tool offering a group communication transcript on demand to find out what was previously discussed in the group conference. Even students who participated from the beginning may benefit from looking up previous parts of the discussion to remind themselves of arguments or decisions already made. This may help conflict resolution. Finally, students may use the group communication transcript to assess their communication, coordination, and collaboration capabilities during autonomous or prescribed reflection and debriefing.

Teachers who are interested in assessing the quality of the collaboration process of a group may benefit from such a tool offering a group communication transcript on demand, so that they can access the group's communication transcript, assess their performance and, if problems (e.g. conflicts) are recognized, may provide help to the group. Finally, teachers may use the group communication transcript in a debriefing with the respective group or for assessing and improving the quality of their group tasks and instructions.

Researchers interested in studying research questions related to group communication may benefit from a tool offering access to group communication transcripts of many groups, provided that data privacy regulations are acknowledged.

Figure 1 shows the lifecycle of a group conference room to be used by a group of students during a course. The lifecycle begins with the creation of a conference room. This conference room can be joined by other members of the group (and potentially the teacher), leading to the establishment of a session. As soon as the desired number of users has joined the conference room and thus the session, the group discussion can begin. During group discussion in the current session, the need for accessing the communication transcript may occur. Thus, any user in the session can download the group communication transcript of the current session at any time. This transcript is then available either for individual reflection or group reflection and can also be used in the group discussion. After the group discussion in the current session has ended, the conference room can be left. If the last user has left the conference room, the current session may be terminated, so that the respective resources are freed. However, the conference room is still available, thus any group member can rejoin the conference room or download a group communication transcript of the last session or of any or all sessions, which happened in the conference room so far, at any time. The downloaded transcript can be used for individual reflection or employed in another session for a group reflection. Deleting the conference room when no further conferences or access to transcripts are needed marks the end of the lifecycle.

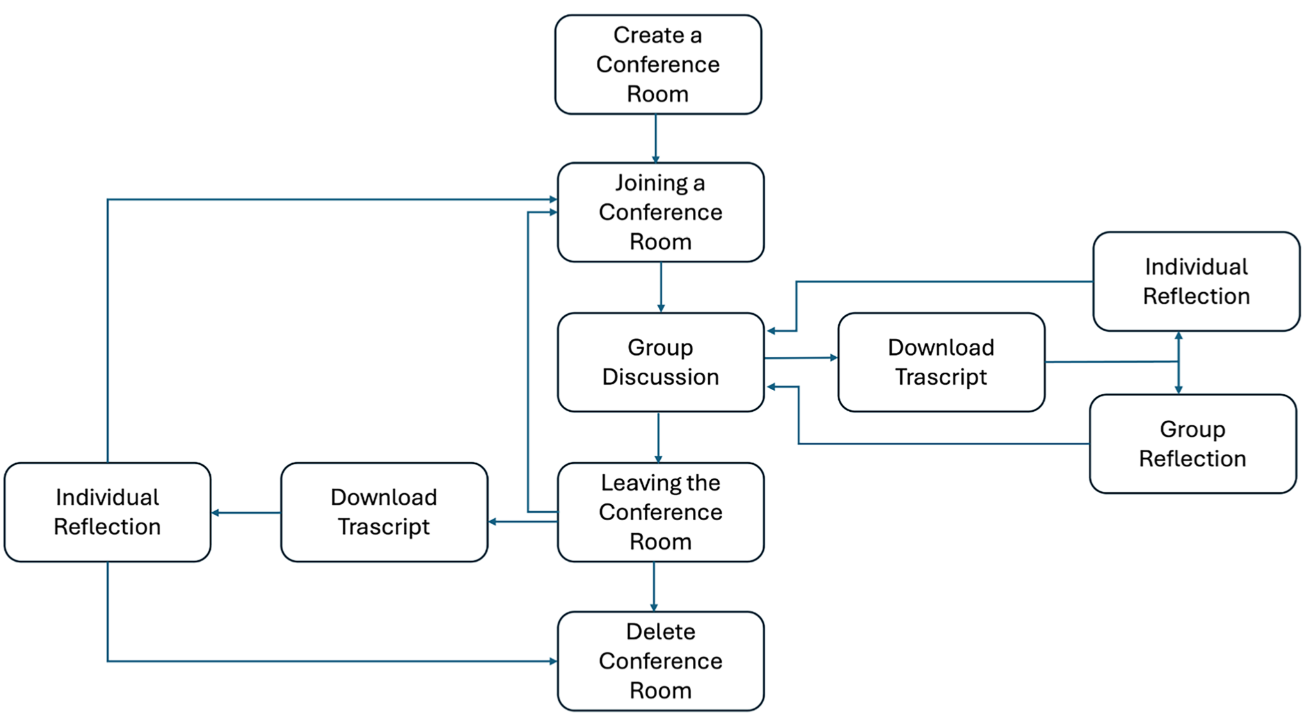

Fig. 1 Activities during the conference lifecycle

3 Requirements

From the educational setting and interests of the stakeholders discussed above and taking the life cycle of a group conference room (cf. Figure 1) into account, we identified the following requirements for a system supporting synchronous audio/video conferences with on-demand access to group communication transcripts:

In educational institutions with large student populations, cost per user is critical, the system should avoid high license cost, suggesting the use of open-source or free software solutions (R1: Avoiding high cost).

A minimal audio/video conferencing system must support real-time transcription of the group conversation for retrospective review by students and teachers, and fulfill the following functional requirements, which are derived from user stories needed to perform the meeting lifecycle (i.e. initiating, joining/leaving, deleting, real-time communication, live transcription and reflection).

In an educational institution, teachers either form groups within the LMS for planned learning groups or permit students to form their own learning groups, e.g. for self-organized learning. To support creation of conference rooms for each group, the system must allow teachers to create a dedicated conference room with a unique identifier that can be used for later joining the conference (R2: Conference Room Creation by teachers). To support self-organized learning groups, the system should allow students to create rooms and invite other members of their group using the unique conference room identifier (R3: Conference Room Creation by students).

The system must permit users of the LMS to join a conference using the identifier (R4: Session Join Capability) and to leave the conference (R5: Session Leave Capability). To reduce system load and respect educational institutions' data protection, the system must allow room creators and teachers to delete conference rooms (R6: Room Deletion).

To enable real-time communication and collaboration, the system must provide users with bidirectional audio (R7: Audio Communication), bidirectional video streaming (R8: Video Communication), and application sharing (R9: Application Sharing).

To enable within- and post-discussion reflection, the system must allow users at any time to download the entire transcript since creation of the conference room or a specified interval (such as the current session) (R10: Transcript Download at any time).

To support different user groups (students, teachers, and researchers) and their use of the transcript, the system must offer different forms of transcripts tailored to the respective user and purpose. For example, latecomers can be addressed with short summaries, while reflections can be conducted on entire transcripts (R11: Transcript Diversity).

To enable live transcription the system must provide real-time audio/video stream ingestion avoiding long delays between demanding and delivering the transcript (R12: Real-Time Stream Processing).

To produce text from audio streams, the system must provide speech-to-text conversion (R13: Speech-to-Text Translation).

To enable the creation of a group discussion transcript, the system must provide timestamped, speaker-labeled transcript storage for reproducibility (R14: timestamped, speaker-labeled Transcript Storage).

To support learning groups employing different languages, the system must support multilingual speech to text translation (R15: Multilingualism).

To support within-discussion and post-discussion reflection, the system must allow individual users to review downloaded transcripts or summaries (e.g., via a dedicated viewer) for individual reflection (R16: Individual Transcript Review). For collaborative reflection, the system must support group discussions related to the transcript (R17: Collaborative Reflection).

Finally, a number of non-functional system quality requirements are derived from the educational setting where three key aspects must be addressed: security, accessibility, and usability. First, all communication channels must be technically secure through end-to-end encryption to guarantee data privacy as requested by data privacy regulations such as the GDPR (R18: Data Privacy). Second, the system should support cross-device compatibility (e.g., desktops, tablets, and smartphones) to enable widespread access by a diverse student community (R19: Multi-Device Support). Finally, the user interface must be intuitive and user-friendly to minimize learning curves and ensure ease of use (R20: Ease of Use), as students should use their effort for solving the group task.

4 State of the art

Audio/video conferencing systems receive and forward both the video stream of a camera (webcam, smartphone cam, device screen) and the audio stream of a microphone or the internal sound of a device. The transmission of audio streams via a browser has already been described by Tu & Loizou [22]. A socket connection was created via a browser to a server on which a speech recognition server processed the audio streams.

A more modern and user-friendly solution for establishing a video chat connection is the WebRTC protocol [23, 24]. This protocol enables users to establish a secure, encrypted peer-to-peer (P2P) connection without requiring additional software, only a browser with internet access is needed. WebRTC supports not only video and audio streams but also canvas content transmission and real-time data channels for auxiliary formats (e.g., chat messages).

However, WebRTC's P2P architecture requires an initial signaling phase to exchange session descriptions (offer/answer) and network metadata (ICE candidates) between peers. While the actual media transmission occurs directly between endpoints, this signaling relies on an intermediary server with a fixed address (e.g., a WebSocket or SIP server). This server facilitates discovery and negotiation but does not participate in the subsequent media relay, preserving WebRTC's end-to-end encryption and P2P efficiency [25].

Available audio/video conferencing systems like Skype, Oovoo, Zoom and Google Meet fulfill the requirements on session management R2—R 6 and audio/video communication functionality (R7—R9) but do not provide the required transcription functionality (R10—R17. An exception is Microsoft Teams [19], which supports transcripts that are available after the end of a meeting but restricts use to holders of an expensive license and can only be used within an organization, which violates R1. It also requires the use of the same language throughout the meeting, violating R15.

Available transcription systems (e.g., ASR systems) do not support session management and audio/video communication (R2—R9) but support the transcription of individual speaker's audio into a transcript (R11, R12, R13, R15) [26]. However, these systems cannot produce a group transcript from a number of individual speaker transcripts (R10, R14) and therefore cannot support group reflection (R16, R17). A typical ASR system consists of four parts: signal processing and feature extraction, acoustic model, language model, and hypothesis search. These four components ensure that the incoming audio signal is cleaned up, filtered for salient feature vectors, weighted, and hypothesized for previously trained words [26]. Leow et al. [27] was able to implement a transcription server tool that receives an audio stream and processes it by using a voice activation detection (VAD) and the Kaldi ASR toolkit with a pre-trained ASR model. Low latency was achieved in audio processing with a comparably low word error rate of 6.45%. Multilingualism (R15) is not fulfilled as it only supports Japanese. An example for a multilingual model with a low word error rate is Whisper [28] trained by OpenAI. This model supports many languages in high quality and includes important functions such as filtering of non-language and conversion of different data types and timestamps. But it does not provide a user interface, real-time transcription, and audio recording.

Overall, the biggest shortcoming of current approaches is the lack of ability to create a group discussion transcript that is available at any time and to any group member (R10) for reflection on the group discussion (R16, R17). Either a non-personalized transcript is created that does not include all information about the discussion (violating R14) and is not equally available to all group members (violating R10), or an audio/video conferencing tool is provided that is not capable of creating a transcript or conversation summary for reflection.

In the following, we present an approach that addresses all requirements and provides group members and teachers with group communication transcripts on-demand at any time.

5 Design

T^3 is a system that combines a virtual group meeting platform with the ability to process audio streams of all meeting participants and save them as text capturing the group discussion by showing speaker labels, timestamps and respective content of the speaker's audio contribution (R14). It is designed to support group projects by providing students and teachers with a web application (R19) for creating conference rooms (R2, R3), joining and leaving synchronous audio/video communication sessions (R4, R5, R7, R8), and, with its integrated transcription and download function (R10, R13), also relieves them of the task of creating notes or meeting minutes (R20). It is an open-source system that can be self-hosted by the university (R1).

5.1 Architecture

Figure 2 shows the architecture of T^3 with its individual components and their connections to each other. Gray boxes denote individual components, blue arrows denote data streams and orange dotted boxes denote subsystems consisting of their components.

T^3 integrates an audio/video communication subsystem and a transcription subsystem. The audio/video communication component operates in conjunction with a session management component, a centralized service that coordinates peer connections and maintains session state. It also allows you to create and delete conference rooms (R2, R3, R6). The session manager dynamically tracks all participants within a conference room (identified by its unique room ID) and manages real-time updates across clients. When a new user (identified by a user ID) joins a session associated with a conference room (R4), the session manager provides them with the current list of active participants and initiates the WebRTC signaling process. This involves exchanging session descriptions (offer/answer) and ICE candidates via WebSocket connections to establish direct peer-to-peer (P2P) audio/video channels with all existing users (R7, R8, R9).

Beyond initial signaling, the session manager broadcasts state changes to all clients in real-time, notifying them when a user joins or leaves the session (R4, R5) or toggles media streams (e.g., enabling/disabling their camera, microphone, or screen-sharing) (R7, R8, R9). While the session manager facilitates signaling and session metadata, it does

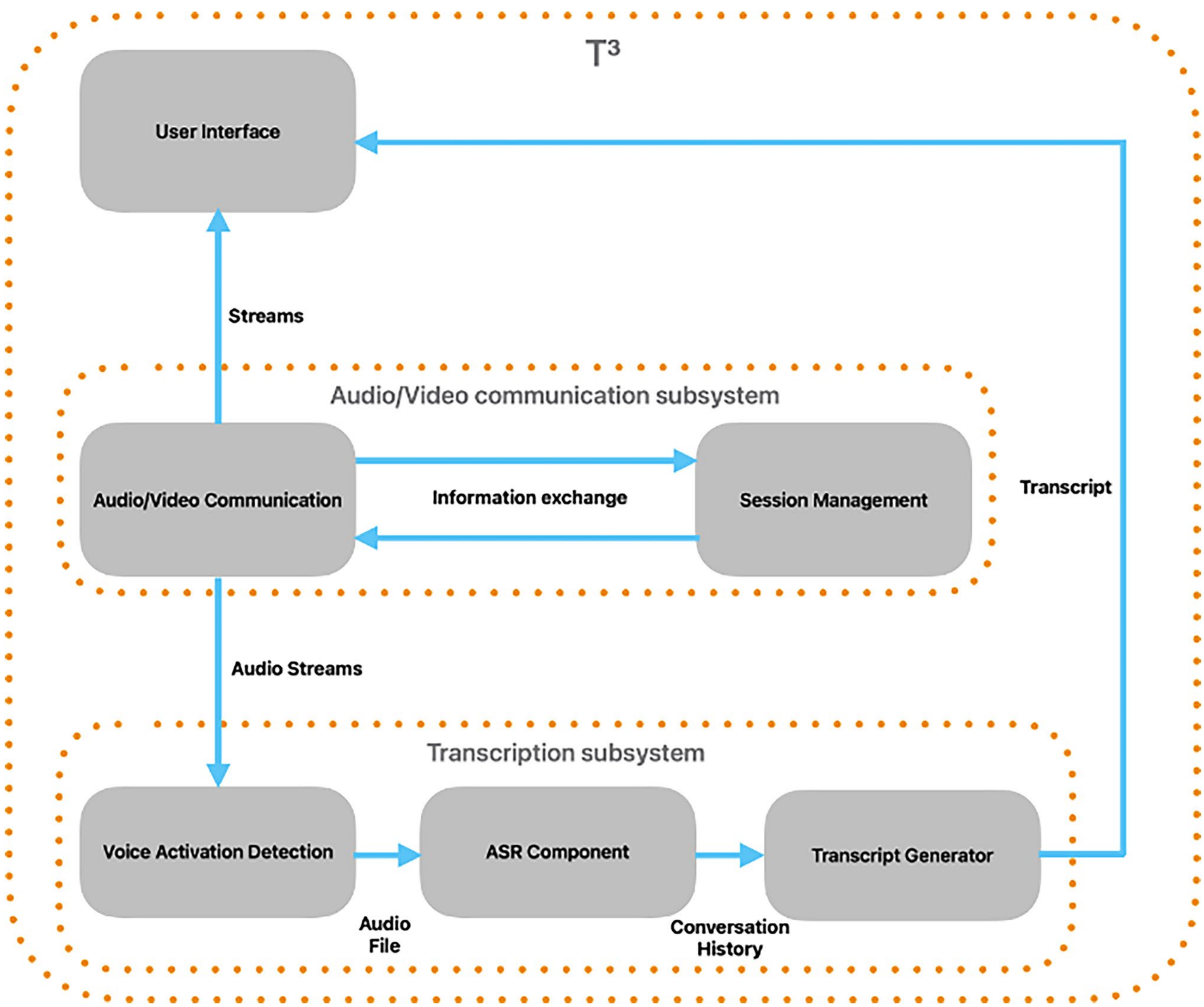

Fig. 2 Architecture of the audio/video conference system with transcribing capability

not handle the actual media transmission, ensuring that all audio/video data streams remain end-to-end encrypted and transmitted directly between peers via WebRTC for optimal performance and privacy (R18).

The audio/video communication component receives these media streams from other participants' audio/video communication components and forwards them to the user interface, where they are decoded and rendered. By decoupling signaling (managed server-side by the session management component) from media transmission (handled P2P), the system maintains low latency and scalability without compromising security.

The transcription subsystem is directly connected to the audio/video communication subsystem and operates through a pipeline of three integrated components. At the core of this subsystem is the Voice Activity Detection (VAD) component, which employs an energy-based algorithm to reliably distinguish between speech segments and silence or background noise in real-time audio streams (R12). This sophisticated component receives raw audio data along with essential client metadata, including user ID and room ID, from the audio/video communication subsystem.

The VAD process begins by conditioning the incoming audio signal through amplitude normalization and application of a band-pass filter (300–3400 Hz) to adjust the frequency range for human speech characteristics. The system then analyzes the processed audio through continuous calculation of short-term energy levels and zero-crossing rates, comparing these metrics against dynamically adjusted thresholds to make robust speech/non-speech determinations. When the algorithm identifies positive voice activity, it initiates a buffering process that captures complete utterances by including configurable silence padding before and after each speech segment. These validated audio segments are encoded in a standardized 16-bit PCM format at 16 kHz sampling rate for optimal compatibility with downstream processing [29, 30].

The processed audio files, now cleanly segmented and properly formatted, are forwarded to the multilingual Automatic Speech Recognition (ASR) component (R13, R15). Here, the system applies language-specific acoustic models to perform accurate speech-to-text conversion (R15). The transcription subsystem maintains strict synchronization throughout this process by preserving the original client metadata and appending precise millisecond-level timestamps to each text segment (R14). These comprehensive data packets, containing the transcribed text along with all relevant contextual information, are then delivered to the transcript generator for final processing. The entire pipeline operates with high concurrency, efficiently handling multiple meeting participants simultaneously while ensuring data integrity through careful management of room IDs and temporal references.

The transcript generator component processes all conversations by combining individual transcripts into a unified group history, organized by client information (user ID, room ID) and timestamp data (R13, R14). This component supports multiple output formats for individual review or collaborative reflection (R10, R11, R16, R17): (1) a complete chronological transcript, (2) a structured data table containing all transcript metadata, (3) when explicitly requested by the user, AI-generated summaries and (4) summary cards through our local implementation of the Llama3 model [31].

The transcript generator employs a conditional execution architecture where the resource-intensive LLM processing is only initiated for summary generation or summary card creation. For standard transcript outputs or data table generation, the transcript generator bypasses the LLM entirely, instead performing deterministic text organization and formatting operations. When activated, the Llama3 implementation processes the complete conversation history through a dedicated pipeline that includes context window management, prompt templating for meeting-specific summarization, and output validation to ensure factual consistency with the source transcripts.

The context window management handles long-form dialogues by using chunking [32] to dynamically segment the input text, ensuring optimal context retention while respecting Llama3's token limits of 8 k token.

The prompt templating contains different system prompts and user prompt templates that include instructions depending on user requirements for individual tasks (R11).

Output validation is performed using NLP keyword extraction of the input and a lemma comparison with the output [33]. For prompts that specify a specific output format, the output is validated against these structures.

All generated outputs, whether LLM-processed or conventional, are delivered through the UI with appropriate metadata labeling to distinguish between verbatim transcripts and AI-generated content.

5.2 UI design

The T^3 UI offers modern application-oriented minimalistic functions such as activating and deactivating the webcam and microphone as well as the option to share the device screen with the other group members in the same room/session (Fig. 3). When accessing the web application (R19), the user is automatically connected to all other users in the same room/session and showed in the UI.

Fig. 3 User interface of T³

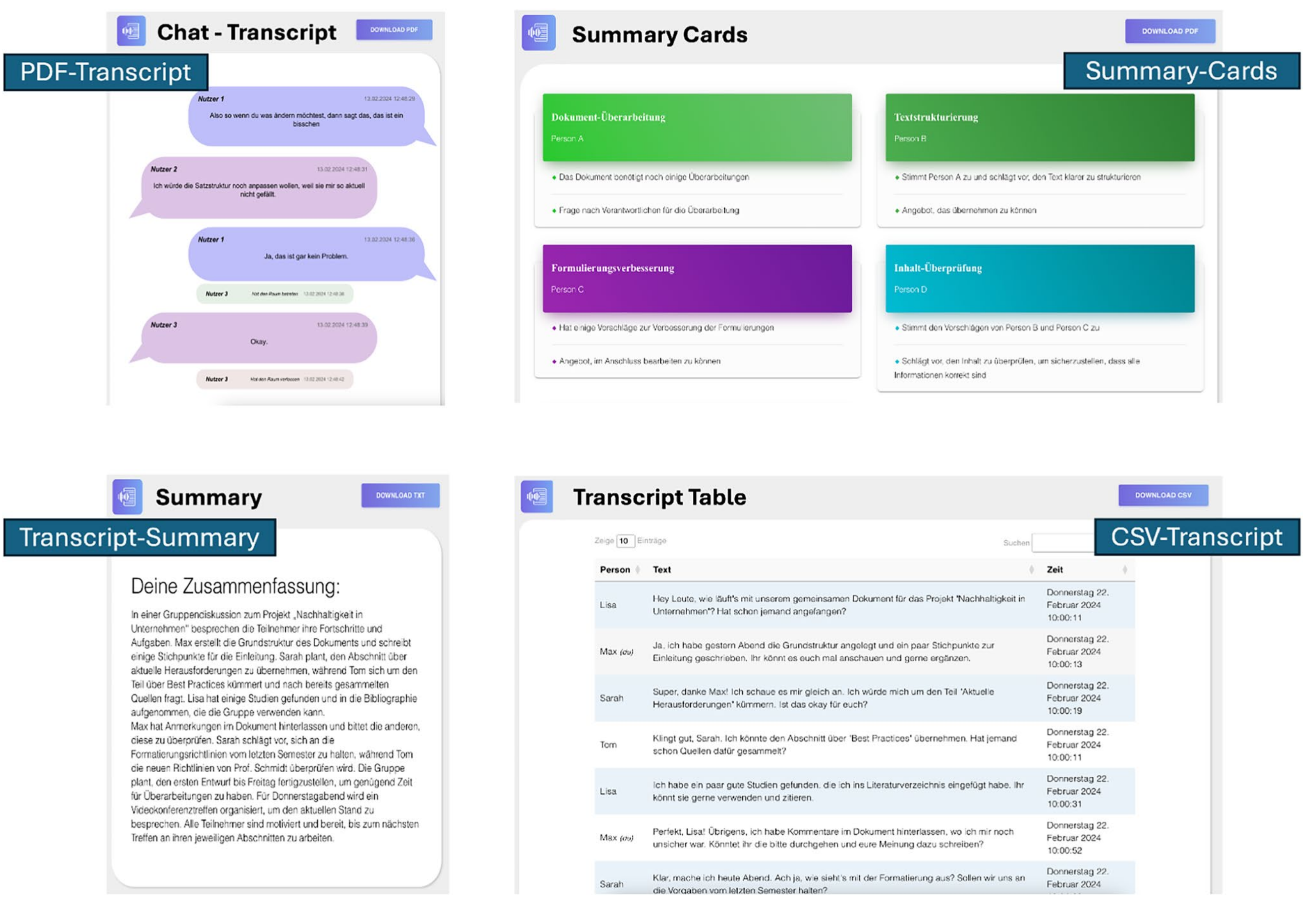

Fig. 4 Possible dialog protocol output formats

The UI provides a dedicated preview component where users can examine their transcript or summary before downloading (R10). As illustrated in Fig. 4, four distinct output formats are available (R11):

5.2.1 PDF-transcript

Designed with a messenger-style interface, this format displays the conversation chronologically, including speaker identifiers, transcribed content, and precise timestamps. It also displays participant join/leave events. A personalization feature aligns the current user's speech bubbles on the right side while other participants appear on the left, mimicking common chat application layouts.

5.2.2 CSV-transcript

The CSV- Transcript is a machine-readable table format which organizes transcript data with columns for speaker identity (annotating the current user with "(you)"), transcribed text, timestamps, and participant join and leave events. The interface allows dynamic sorting (ascending/descending by any column) and content filtering via search terms, with the default view showing entries in reverse chronological order.

5.2.3 Transcript-summary

The Transcript-Summary is a condensed text overview generated by the Llama3 LLM, distilling key conversation points while preserving contextual accuracy. This output targets quick content review without temporal or speaker-specific details.

5.2.4 Summary-cards

The Summary-Cards are also LLM-generated (Llama3) and are an innovative format structuring the conversation into digestible units. Each card represents a distinct discussion topic, with associated subtasks and participant contributions clearly tagged, enabling rapid visual scanning and information retrieval.

All formats are accessible through a unified download interface, with the LLM-dependent options (3–4) clearly labeled as AI-generated content. The design prioritizes both human readability (PDF) and computational usability (CSV), while the summary variants cater to different review scenarios.

6 Implementation

To keep the application as simple as possible and ensure compatibility with the user's device regardless of the operating system, without requiring additional software, the browser was used as the client tool for deploying T^3 as a web application (R19, R20).

T^3 consists of a ReactJS [34] client-side server and a Python backend server. The ReactJS server hosts the user interface as a web application (Fig. 2), forwarding user media streams to the session management component and the transcription subsystem. The Python server hosts the session management component (signaling server) and the transcription subsystem (VAD, ASR, and transcript generator), which handles transcription. The transcription interface operates in the background, invisible to the user.

The web application is accessed via a browser and a specific URL. The "ID" parameter in the URL specifies the virtual group room to join (R4); if omitted, a new room is created with a UUID (R2, R3). Upon entry, the user grants permission for audio/video stream access via the MediaDevices API. A WebSocket connection is then established (via NodeJS) to the Python signaling server, transmitting stream statuses (muted/active) and the room ID. A user ID is returned, and if additional users join with the same room ID, WebRTC offers/answers are exchanged via the signaling service, creating peer-to-peer (P2P) connections for each user. Audio/video streams are then rendered on the webpage (R7, R8).

Communication is performed using a full-mesh P2P topology, reducing server dependency and maintaining persistent connections even during server failures. While this approach increases CPU and network load (requiring one stream per participant) compared to MCU or SFU architectures [35], the pure P2P design was prioritized for its cost-effectiveness on university hardware and superior scalability for small groups.

For data persistence, T^3 employs an SQLite database (stored locally on the server) to manage transcripts and metadata. SQLite was chosen for its lightweight, serverless architecture, which aligns with the goal of minimizing dependencies while ensuring reliable storage. The database schema includes tables for user IDs, timestamps, room IDs, and transcribed text, enabling efficient retrieval for generating conversation histories or summaries. For scalability, SQLite's file-based storage was chosen as it efficiently handles small-to-medium workloads, though future versions may transition to a distributed database such as PostgreSQL to accommodate larger-scale deployments.

Concurrently with signaling, a socket connection to the transcription subsystem is used to submit user data and a WebRTC offer. The transcription subsystem responds with a WebRTC answer, creating a P2P connection for audio transmission. The incoming stream is formatted and analyzed by a VAD [30]; voice activity triggers recording to a.wav file. These files are sent to OpenAI's Whisper ASR [28] using a multilingual model for transcription (R15). The output text, along with user ID, timestamp, and room ID, is stored in the SQLite database. This process runs in parallel for all users (R12, R13, R14).

Users can download a chronological conversation history or summary (Fig. 4) via a click on the button (see Fig. 3) which triggers the transcript generator using an SQL query and LLM prompting if needed, to provide the chosen output (R10). The output is downloadable as PDF, CSV, or TXT, providing a local copy. By providing a local copy of the transcript and summary, users can reflect on the course of the conversation and its content (R16). It is also be conceivable to share the transcript with the other participants in the group using the application/screen share functionality of the audio/video communication subsystem (R17).

To ensure privacy, sessions are automatically deleted when rooms become empty. Temporary.wav files generated during voice activity detection are deleted after transcription, while the resulting transcripts persist in the database until manually purged by deleting the conference room (R6, R10).

Session termination occurs by closing the browser tab or logging out, severing all connections. Empty sessions are automatically deleted by the session manager for data protection (R18). Since all data is stored on university servers, data privacy can be ensured (R18).

7 Evaluation

To evaluate acceptance and applicability, we interviewed four experienced computer science teachers. Their feedback confirmed that both aspects were deemed appropriate, comprehensive, and useful.

To test the functionality and usability of the system, an experimental environment was set up in which test persons were to use T^3. T^3 was installed as a service on a Linux server and used with the Whisper "medium" model. The operating system Linux was chosen for practical reasons because it is an open-source and cost-free OS, which is widely used in academic context. Whisper medium was preferred over whisper large because of less resource needs (only half resources needed compared to Whisper large), speed (twice as fast compared to Whisper large) and only slightly worse WER (for German) which is still better than human WER [28]. Especially speed is important since T^3 aims at near real-time solution.

To demonstrate the functionality of T^3, testing T^3 with a group of 4 people is deemed sufficient. The group consisted of 2 male and 2 female students of different semesters and disciplines, aging between 27 and 31, with German as a mother tongue. It was given the task of discussing a scientific article previously read, followed by a debriefing using the generated PDF transcript regarding the suitability of the collaboration process to solve the task. The functionality test followed no script or guidelines beyond the scientific article, ensuring it remained as realistic as possible. Individual semi-structured interviews were then conducted regarding the collaboration and technical difficulties to capture the subjects' impressions.

During the tests, users had no problems creating, entering and leaving conference room. Peer-to-peer connections between the users' clients were established without any problems. There were also no technical limitations when people interacted with each other, allowing the group to successfully complete the task. The transcription and creation of interview transcripts were successfully completed, and a personalized PDF transcript was created for each user when they clicked on the download button (Fig. 3) during and after the conference. The test subjects did not notice any negative impact on their collaboration process as a result of the automatic recording of the audio tracks.

The evaluation of the transcript and the determination of the word error rate (WER) were carried out by an expert who had access to both the recording of the experiment and the resulting transcript. In order to compute the WER the expert transcribed the audio files of the recording manually. This manual transcript was than compared by the same expert with the automatically generated transcript. For this purpose, the automatically generated transcript was normalized by filtering out vocal fillers (ah, uh, hmm, mhm, uh) as well as converting numbers to the written form.

For every sentence in the manual transcript, the expert checked for an identical sentence in the automatic transcript. If differences occurred the expert looked for substitutions (spelling differences), missing words or added words. The percentage of the WER was then computed by dividing the sum of substitutions, missing and added words by the total number of words from the manual transcript. By doing so a WER of 8.04% was measured. The conversations were in German and have a similar value to the word error rates shown by Whisper medium (for example COMMON VOICE 9 German with 8.5% WER and MULTILINGUAL LIBRISPEECH German with 7.4% WER) [28]. This value is almost as high as that of a professional human transcriber, who has a WER between 5.0 and 10.5% [28, 36, 37].

8 Conclusion

To support reflection about the group work process and communication in distributed synchronous meeting, all members of a distributed learning group should have access to an up-to-date group discussion transcript on demand. For the development of such an audio/video conferencing system, two research questions were answered in this paper: The requirements of such a tool regarding the necessary use cases were identified (RQ1) based on a new process model describing the activities of group members, teachers and researchers before, during and after conferences during group learning tasks. This process model was used to specify 20 requirements on a system supporting synchronous audio/video conferences with on-demand access to group communication transcripts, which supports the activities needed for using a group communication transcript to help solving the task, assess and improve the collaboration process, solve conflicts, provide support for reflection, and analyze group communication. A conceptual architecture of an audio/video conferencing tool with on-demand transcription capability consisting of an audio/video communication subsystem and a transcription subsystem was introduced (RQ2). The implementation of this architecture using open-source components was demonstrated by the T^3 tool. An evaluation showed functional correctness and usability of the tool. T^3 was tested in an experimental environment and showed a good WER of 8.04%. This value is within the WER range of a professional human transcriber of 5.0–10.5% [28, 36, 37] and is therefore perfectly suitable for reflecting on group discussions. A similar value is also found in the paper by Radford et al. [28] and Gao et al. [38], which suggests that WebRTC transmission and VAD filtering do not have a negative impact on transcription quality. It can therefore be assumed that other languages will have similar WER values to those described by Radford et al. [28]. Nevertheless, further tests should be conducted in this regard.

The approach goes beyond current audio/video conferencing systems and transcription support with respect to its open-source nature and the ability to create a multilingual group discussion transcript that is available at any time and to any group member for within-meeting and post-meeting reflection on the group discussion.

T^3 will be used in university courses to collect data from students and make it available to them for improving their learning and collaboration capabilities. As the system has not yet been actively used in university courses, its effects on learning and improvement of collaboration processes are not yet known. However, research by Hwang et al. [15], Kuo et al. [16], and Huang et al. [17] suggests that systems such as T^3 may have a positive impact on student learning outcomes. The reader should note that on-demand multilingual transcription may not only be beneficial for reflection in learning groups but may also benefit collaborating groups in a work context. The paper is therefore not only relevant for teachers in schools and universities to provide their students with a platform for reflecting on group discussions, but also for researchers to collect data and analyze the results from many groups within a learning institution. It can thus be considered as an enabler of future studies.

While T3 effectively supports collaborative learning through real-time transcription, several limitations must be acknowledged. T^3 is currently limited to textual storage of the spoken word in order to conserve resources and not overload the database. Although Whisper medium achieves an 8.04% WER for German, performance may vary across languages and noisy environments, warranting domain-specific fine-tuning. The evaluation's limited scale ($N = 4$ in a controlled setting) calls for longitudinal classroom studies to assess pedagogical impact. In the future, we plan to employ Whisper large turbo which provides a speedup factor of 4 compared to Whisper medium with slightly better WER than Whisper medium and only slightly worse WER than Whisper large. We also plan to perform a larger study involving users with different accents, native as well as non-native speakers in a university setting, as these factors will significantly affect performance as shown by Wills et al. [39].

Author contribution TK developed the tool and analyzed the data (transcription WER). Both (TK, JH) interpreted the results and were major contributors in writing the manuscript. Both authors read and approved the final manuscript.

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Data availability Datasets used and analyzed during the current study are available from the corresponding author on reasonable request. Due to the sensitive nature of the data and to ensure participant confidentiality, the raw data supporting this study cannot be made openly available. Data access may be considered on a case-by-case basis for qualified researchers, subject to institutional review board (IRB) approval and a formal data-sharing agreement. Requests should be directed to Thomas Kasakowskij at Thomas.Kasakowskij@Fernuni-hagen.de.

Declarations

Ethics approval and consent to participate This study, involving human research participants, was conducted in strict adherence to ethical principles. The research protocol was approved by the Ethics Committee of Fernuniversität in Hagen, in accordance with the Guiding Principles for Ethical Research at Fernuniversität in Hagen. The study was carried out in accordance with the guidelines and regulations set forth by this ethics committee, ensuring the protection of the rights and welfare of all participants involved.

Competing interests The authors declare no competing interests.

Informed consent The authors confirm that informed consents, encompassing both Consent to Participate and Consent to Publish, have been duly obtained from all adult participants involved in this study. As there are no participants under the age of 18, no consent was required from parents or legal guardians. All participants were fully informed about the study's objectives, procedures, potential risks, and benefits, and they voluntarily agreed to participate and to the publication of the research outcomes. This confirmation is made with the full knowledge and agreement of all authors.

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